

PREMARKET APPROVAL APPLICATION (PMA)

NeuroStim Deep Brain Stimulation System

NeuroTech Medical, Inc.

Submission Date: 10/29/2025
Device Classification: Class III

MODULE 5: CLINICAL STUDIES

5.1 Pivotal Clinical Trial

Study Design: Multicenter, prospective, double-blind, randomized controlled trial

Study Population: 298 subjects with Parkinson's disease

Primary Endpoint: Change in UPDRS Part III motor score at 12 months

Study Duration: 24 months follow-up

5.2 Clinical Results

Primary Endpoint Results:

- Treatment Group: 52% improvement in UPDRS III ($p < 0.001$)
- Control Group: 8% improvement in UPDRS III
- Difference: 44% (95% CI: 38-50%)

Secondary Endpoints:

- Quality of Life (PDQ-39): 38% improvement
- Activities of Daily Living: 45% improvement
- Medication Reduction: 50% reduction in L-DOPA equivalent

5.3 Safety Results

Adverse Events:

- Serious Adverse Events: 12% of subjects
- Device-Related SAEs: 4% of subjects
- Most Common AEs: Mild dyskinesia (18%), mild dysarthria (12%)
- No unanticipated adverse device effects